Gene, 74 (1988) 365–373
Elsevier

GEN 02747

An *Escherichia coli* vector to express and purify foreign proteins by fusion to and separation from maltose-binding protein

(Recombinant DNA; *malE*; factor X_a protease; paramyosin; crosslinked amylose; β-galactosidase; *Dirofilaria immitis*; affinity purification; over expression; phage λ vectors)

Claude V. Maina, Paul D. Riggs, Andres G. Grandea III, Barton E. Slatko, Laurie S. Moran, John A. Tagliamonte, Larry A. McReynolds and Chu di Guan *

New England Biolabs, 32 Tozer Road, Beverly, MA 01915 (U.S.A.)

Received 7 June 1988
Revised 26 July 1988
Accepted 27 July 1988
Received by publisher 6 August 1988

SUMMARY

A plasmid vector has been constructed that directs the synthesis of high levels (approximately 2% of total cellular protein) of fusions between a target protein and maltose-binding protein (MBP) in *Escherichia coli*. The MBP domain is used to purify the fusion protein in a one step procedure by affinity chromatography to crosslinked amylose resin. The fusion protein contains the recognition sequence (Ile-Glu-Gly-Arg) for blood coagulation factor X_a protease between the two domains. Cleavage by factor X_a separates the two domains and the target protein domain can then be purified away from the MBP domain by repeating the affinity chromatography step. A prokaryotic (β-galactosidase) and a eukaryotic (paramyosin) protein have been successfully purified by this method.

INTRODUCTION

Gene fusions have been used successfully to express and purify proteins produced from recombinant DNA molecules (Silhavy et al., 1984) and have been made for several genes (Vermersch et al., 1986; Young et al., 1985). A common class of fusions involve β-galactosidase whereby the fusion protein is expressed in *Escherichia coli* using the ribosome-binding site and translational start of *lacZ* (Shuman et al., 1980; Silhavy et al., 1984). A widely used *lacZ* fusion system in *E. coli* is λgt11 (Young and Davis,

Correspondence to: Dr. P.D. Riggs, New England Biolabs, 32 Tozer Road, Beverly, MA 01915 (U.S.A.) Tel. (508)927-5054.
* Present address: Research and Development Department, Shenzhen Science and Industrial Park Corporation, Nantou, Shenzhen, Guangdong, People's Republic of China.

Abbreviations: aa, amino acid(s); Ap, ampicillin; Δ, deletion; factor X_a, activated factor X protease; FPLC, fast protein liquid chromatography; fX, factor X_a recognition sequence; IgG, immunoglobulin G; MBP, maltose-binding protein; nt, nucleotide(s); P_{lac}, *lac* operon promoter; PolIk, Klenow (large) fragment of *E. coli* DNA polymerase I; PAGE, polyacrylamide gel electrophoresis; SDS, sodium dodecyl sulfate; [ ], denotes plasmid-carrier state.

0378-1119/88/$03.50 © 1988 Elsevier Science Publishers B.V. (Biomedical Division)

1983), whereby fragments of genomic DNA or cDNA are cloned at the 3′ end of *lacZ* in a λ phage vector. Such hybrid proteins can be purified by affinity chromatography using anti-β-galactosidase affinity columns. A similar fusion system uses protein A from *Staphylococcus aureus* (Löwenadler et al., 1986; Valerie et al., 1987) or a synthetic IgG-binding domain of protein A (Löwenadler et al., 1987) to purify fusion proteins by binding to an IgG-resin.

An alternative approach has been to fuse target proteins to maltose-binding protein (MBP; Bedouelle and Duplay, 1988; Guan et al., 1988) coded for by *malE* of *E. coli*. This approach uses a crosslinked amylose resin as an affinity matrix to purify the MBP fusion protein and gives a high yield of pure fusion protein in a single purification step, carried out under non-denaturing conditions.

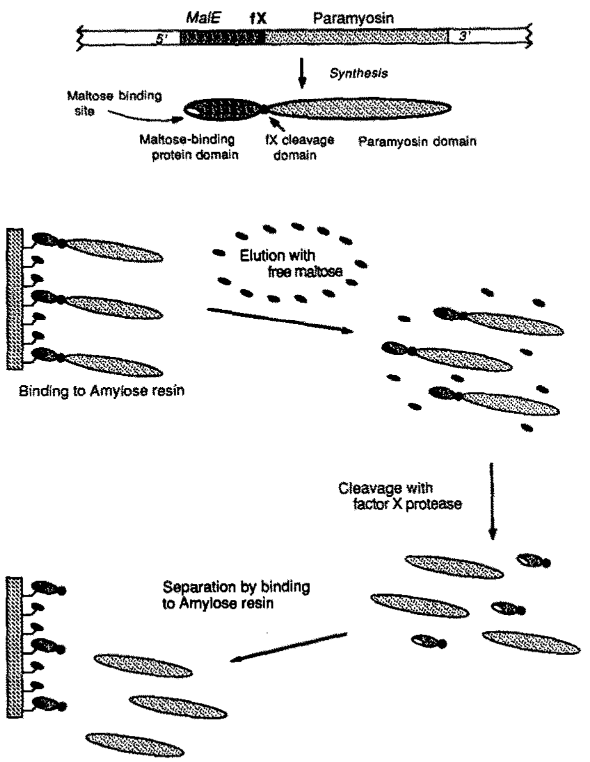

Fig. 1. Schematic representation of the synthesis and purification of MBP fusion protein, with paramyosin as an example. fX denotes the recognition site for factor X_a. The cleared lysate of induced cells is run through a crosslinked amylose column. After washing, the fusion protein is eluted with a dilute maltose solution. The purified fusion protein is then cleaved with factor X_a and the paramyosin domain is purified away from the MBP domain by re-passage through the crosslinked amylose column.

An improvement of this system of expression and purification would allow the separation of the target domain from the MBP domain by a site specific proteolytic cleavage after purification. The resulting end product would be purified target protein. A few methods have been developed to accomplish this (see Germino and Bastia, 1984; Haffey et al., 1987; Nilsson et al., 1985; for examples). We believe the best method uses blood coagulation factor X_a protease (factor X_a). Factor X_a converts prothrombin, by specific cleavage at the tetrapeptide, Ile-Glu-Gly-Arg, to thrombin during the coagulation process (Jackson, 1985). Thus, if the recognition site is placed before the target protein domain in a fusion, factor X_a will cleave specifically at that site releasing the target protein without any additional N-terminal residues. This method has been described previously (Nagai et al., 1985; Nagai and Thøgersen, 1984) and we have adapted it for our system.

In this paper we describe the construction of a MBP fusion vector, pCG806fX. This plasmid contains the DNA sequence coding for the factor X_a protease recognition site downstream of *malE* such that protein fusions constructed with pCG806fX can be cleaved with factor X_a, separating the MBP and target protein domains. The target protein domain can then be purified away from the MBP domain using the same amylose resin to bind the MBP domain while the target protein domain flows through (Fig. 1). We describe examples using a prokaryotic and a eukaryotic target protein.

MATERIALS AND METHODS

(a) *Escherichia coli* strains and culture conditions

All plasmid constructions and protein purifications were done in *E. coli* strain, 71–18, Δ(*lac-proAB*) *thi supE* [F′ *proAB*$^+$ *lacI*^{q} *lacZ*ΔM15] (Yanisch-Perron et al., 1985). The *lacI*^{q}-coded repressor strongly represses the *lac* promoter on plasmids described in this paper and therefore allows cloning of sequences that may be lethal when expressed in even low amounts. Bacteria were cultured as described (Guan et al., 1988).

(b) Construction and analysis of recombinant plasmids

All enzymes used in plasmid constructions and analysis were obtained from New England Biolabs and used as described by the manufacturer. *E. coli* was transformed with ligation mixtures as described (Maniatis et al., 1982). Gel electrophoresis, DNA isolation and gel electroelution of DNA fragments were all performed as described (Maniatis et al., 1982). Double-strand DNA sequencing was performed by a 'collapsed plasmid' protocol (Slatko et al., 1987). Chemical synthesis of oligodeoxynucleotides was performed at New England Biolabs on a Biosearch 8600 automated synthesizer.

(c) Purification of fusion proteins

Expression and purification of fusion proteins, and preparation of crosslinked amylose resin were performed as described (Guan et al., 1988). SDS-PAGE and immuno blots were performed as described (A.G.G., L.K. Tuyen, N. Asikin, T.B. Davis, M. Philipp, C. Cohen, and L.A.McR., in press) using either serum made against adult *D. immitis* worms (Philipp and Davis, 1986) or against purified MBP (gift from P. Bassford, U. of N. Carolina, Chapel Hill, NC, through J. Beckwith, Harvard Medical School, Boston, MA).

(d) Digestion of fusion proteins with factor X_a protease and purification of domains

Factor X_a was a gift from C.M. Jackson, Blood Services Research Laboratory, S.E. Michigan Region, American Red Cross, Detroit, MI. MBP-paramyosin was digested with factor X_a as described by Nagai and Thøgersen (1984). MBP-β-galactosidase was first denatured by dialysis in 8 M urea followed by dialysis in 10 mM Tris·HCl, pH 7.2. The concentration of factor X_a in digestions was 1%, by weight, of the fusion protein to be digested. After cleavage of MBP-paramyosin with factor X_a, the two domains were separated by re-passage over a crosslinked amylose column (see Fig. 5 legend for details).

RESULTS AND DISCUSSION

(a) Construction of maltose-binding protein fusion vector

The construction of pCG806fX is depicted in Fig. 2. The *Stu*I site immediately following the factor X_a recognition site is used to clone target sequences. Since the DNA sequence coding for the factor X_a recognition site reads in the 5′-to-3′ direction along either strand of the *Bam*HI fragment, if the target sequence is inserted in the wrong orientation with respect to *malE*, it can be easily reversed by digestion with *Bam*HI, followed by religation and transformation (Nagai and Thøgersen, 1984; Fig. 2A). pCG806fX places an additional 38 nt between the *malE* and *lacZ* sequences. The result is an out-of-frame *malE-lacZ* fusion and α-complementation cannot be used for screening of insert DNA (see section **f** below).

(b) Construction of *malE*-paramyosin and *malE-lacZ* gene fusions

Paramyosin is a large (97 kDa) protein that, as a dimer, forms the core of thick filaments in invertebrate muscle and shows great similarity across invertebrate species (Cohen et al., 1971; Winkelman, 1976). A 2.6-kb *Eco*RI DNA insert coding for most of the paramyosin gene from *Dirofilaria immitis* (the causative agent of dog heartworm; Knight, 1977; A.G.G., L.K. Tuyen, N. Asikin, T.B. Davis, M. Philipp, C. Cohen, and L.A.McR., in preparation) was subcloned from its λgt11 vector into pCG806fX (Fig. 2C).

The *malE-lacZ* fusion was constructed using the *lacZ* gene of pMLB1034 (Casadaban et al., 1980; Fig. 2C).

(c) Isolation of fusion proteins by affinity chromatography

The MBP-β-galactosidase and MBP-paramyosin fusion proteins were purified using the affinity of the MBP domain to amylose as described previously (Guan et al., 1988). The purification of MBP-paramyosin can be seen in Fig. 3. The gel stained with Coomassie brilliant blue shows three major bands and several minor bands in the fraction that

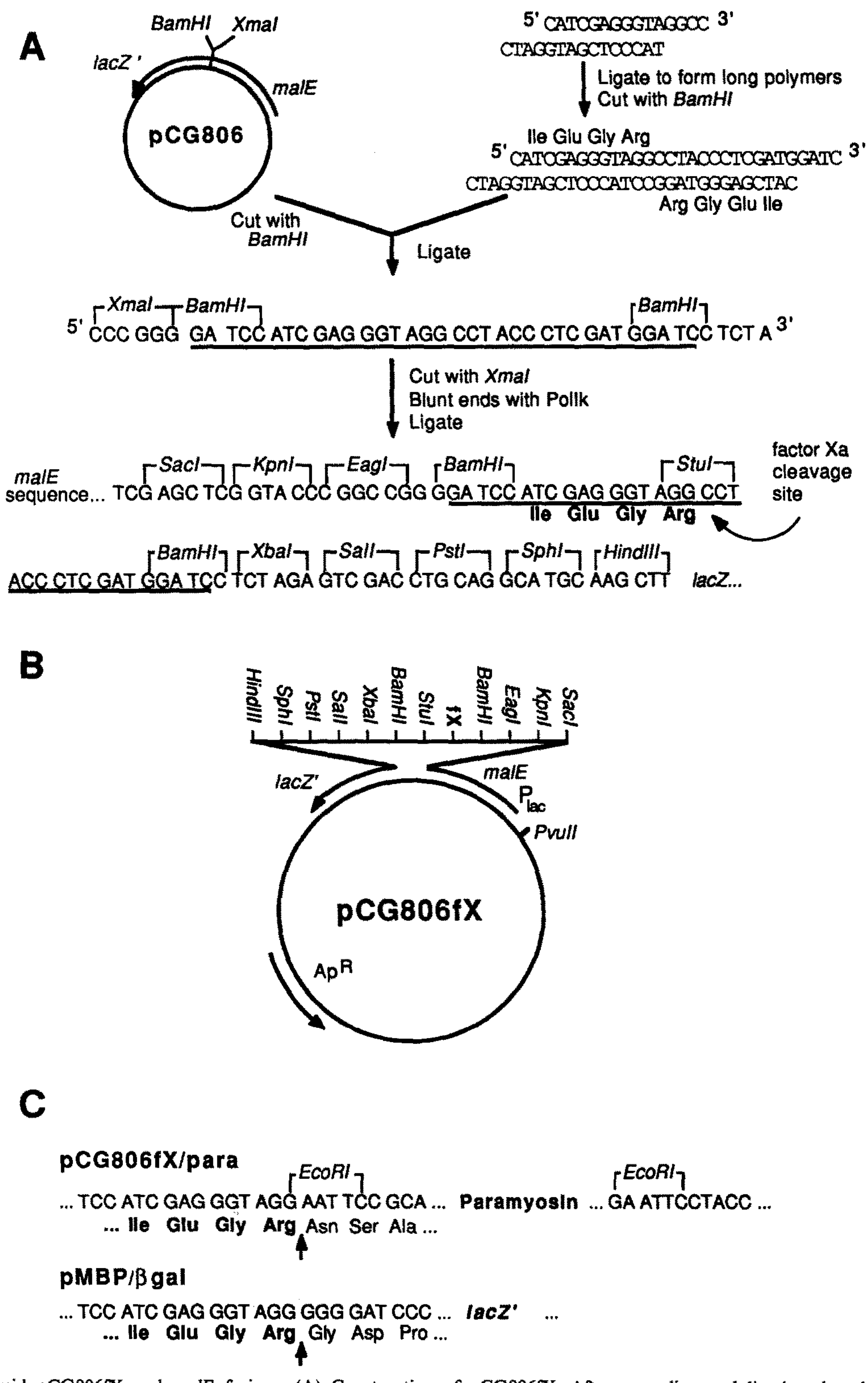

Fig. 2. Plasmid pCG806fX and *malE* fusions. (A) Construction of pCG806fX. After annealing and ligating the phosphorylated oligodeoxynucleotides coding for the factor X_a recognition site (5′-CATCGAGGGTAGGCC-3′ and 3′-CTAGGTAGCTCCCAT-5′), large molecules were size selected by gel electrophoresis, eluted from the gel and cut with *Bam*HI, resulting in a dimer of the annealed polydeoxynucleotide with *Bam*HI sticky ends. This *Bam*HI DNA fragment codes for the 4-aa recognition site for factor X_a (Ile-Glu-Gly-Arg) reading in either direction. This was inserted into the *Bam*HI site of pCG806 (Guan et al., 1988), which lies between the *malE*

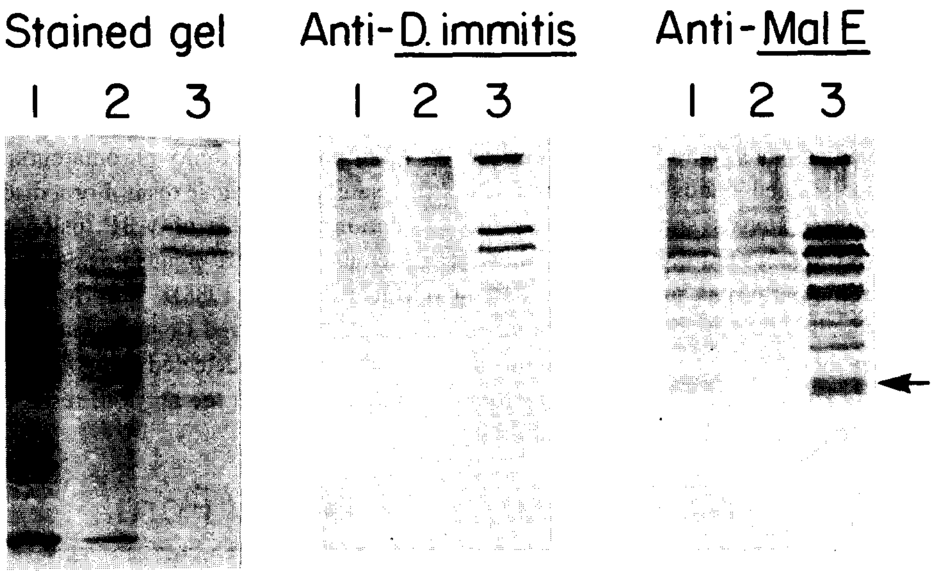

Fig. 3. Purification of MBP-paramyosin fusion protein from *E. coli*. Samples were electrophoresed through 0.1% SDS-10% polyacrylamide gels. Lanes: 1, 10 μg of total cellular proteins from induced cells; 2, 10 μg of flow-through from the crosslinked amylose column; 3, 5 μg of purified fusion protein eluted from the crosslinked amylose column. Left panel, gel stained with Coomassie brilliant blue. Center panel, immuno blot using rabbit anti-*D. immitis* serum. Right panel, immuno blot using anti-MBP serum. After incubation of blots with the primary serum, they were incubated with a secondary serum linked to horseradish peroxidase H and protein bands visualized by the addition of enzyme substrate as described (A.G.G., L.K. Tuyen, N. Asikin, T.B. Davis, M. Philipp, C. Cohen, and L.A.McR., in press). The minor protein bands seen with anti-MBP serum but not anti-*D. immitis* serum are due to the high titre of the anti-MBP serum that allows the detection of small amounts of truncated MBP-paramyosin fusions. These bands are not detected when higher dilutions of the anti-MBP serum are used (data not shown). The reactivity in both the lysate lane and flow-through lanes using anti-*D. immitis* serum and flow-through lane using anti-MBP serum is background that is also seen in cells that were not transformed (data not shown). Arrow points to MBP.

is bound to, and then eluted from the column with maltose (lane 3). The multiple banding pattern is similar to that seen for the same paramyosin sequences expressed from λgt11 (A.G.G., L.K. Tuyen, N. Asikin, T.B. Davis, M. Philipp, C. Cohen, and L.A.McR., in preparation) and can be explained by premature termination of the fusion protein or its digestion by *E. coli* proteases. The lowest of the three major bands is of a similar size to MBP and is either the endogenous MBP or a cleavage product of the fusion protein.

The identity of the eluted proteins was determined

---

and *lacZ'* sequences of the plasmid. Several resulting plasmids containing the *Bam*HI insert were sequenced at the insertion site and one plasmid containing only one copy of the *Bam*HI fragment was used in further constructions. The factor X_a-recognition sequence in this plasmid is now out of frame with the upstream *malE* sequences. To place it in frame with *malE*, the plasmid was cut with *Xma*I; the ends were made blunt with PolIk, and the plasmid recircularized, resulting in an *Eag*I recognition site, to finally create pCG806fX. The inserted oligodeoxynucleotide is underlined and cloning sites are depicted above their sequence. The factor X_a recognition site is shown in bold letters and site of cleavage is shown by arrow. (**B**) Map of pCG806fX. Cloning sites between *malE'* and *lacZ'* are shown above the plasmid. fX denotes the recognition site for factor X_a. (C) DNA and amino acid sequence of the junction region of *malE'* and paramyosin (above) and β-galactosidase (below). The factor X_a recognition site is shown in bold letters and site of cleavage is shown by upward arrow. pCG806fX/para was constructed as follows. The 2.6-kb DNA fragment of λcDi2 (A.G.G., L.K. Tuyen, N. Asikin, T.B. Davis, M. Philipp, C. Cohen, and L.A.McR., in press) was purified away from the λgt11 vector by gel electroelution. The ends of the fragment were then made blunt with PolIk, followed by ligation to *Stu*I-digested pCG806fX. The ligation mixture was used to transform *E. coli* 71–18. Plasmid DNA was isolated from transformants and the presence and orientation of paramyosin insert DNA determined by restriction digestion. pMBP/βgal was constructed as follows. pMLB1034 (Casadaban et al., 1980), was cleaved at the 5′ end of *lacZ* with *Sma*I. A *Pvu*II-*Stu*I fragment from pCG806fX containing the *malE*-fX sequences was inserted into the *Sma*I site by blunt end ligation. The ligation mixture was used to transform *E. coli* 71–18. Plasmid DNA was isolated from transformants and the presence of *malE*-fX insert DNA determined by restriction digestion. All arrows concentric with circles depict extent of indicated genes and direction of transcription.

by immunoblot analysis. Identical gels to Fig. 3, left panel, were transferred to nitrocellulose and reacted with either serum made against adult *D. immitis* worms (center panel) or against purified MBP (right panel). Fig. 3, center panel, lane 3, shows that the upper two major stained bands of the eluate are reactive with anti-*D. immitis* serum; whereas the lower band is not. Fig. 3, right panel, lane 3, shows an identical banding pattern as lane 3 in the stained gel indicating that all of the eluted proteins are either MBP or fusions of MBP. The upper two major bands seen in lane 3 of both the anti-*D. immitis* serum and anti-MBP serum panels demonstrate that these proteins are indeed MBP-paramyosin fusions. The lower band, marked by an arrow, is reactive with anti-MBP serum only demonstrating that it is MBP.

The purification of the fusion protein is a single step process: affinity chromatography of a cleared lysate over amylose resin. For a discussion of purification and yields of MBP fusion proteins and for details of MBP-β-galactosidase purification see Guan et al. (1988). For MBP-paramyosin we generally achieved 1 mg of protein per liter of culture and have not detected any MBP-paramyosin that did not bind to the amylose resin.

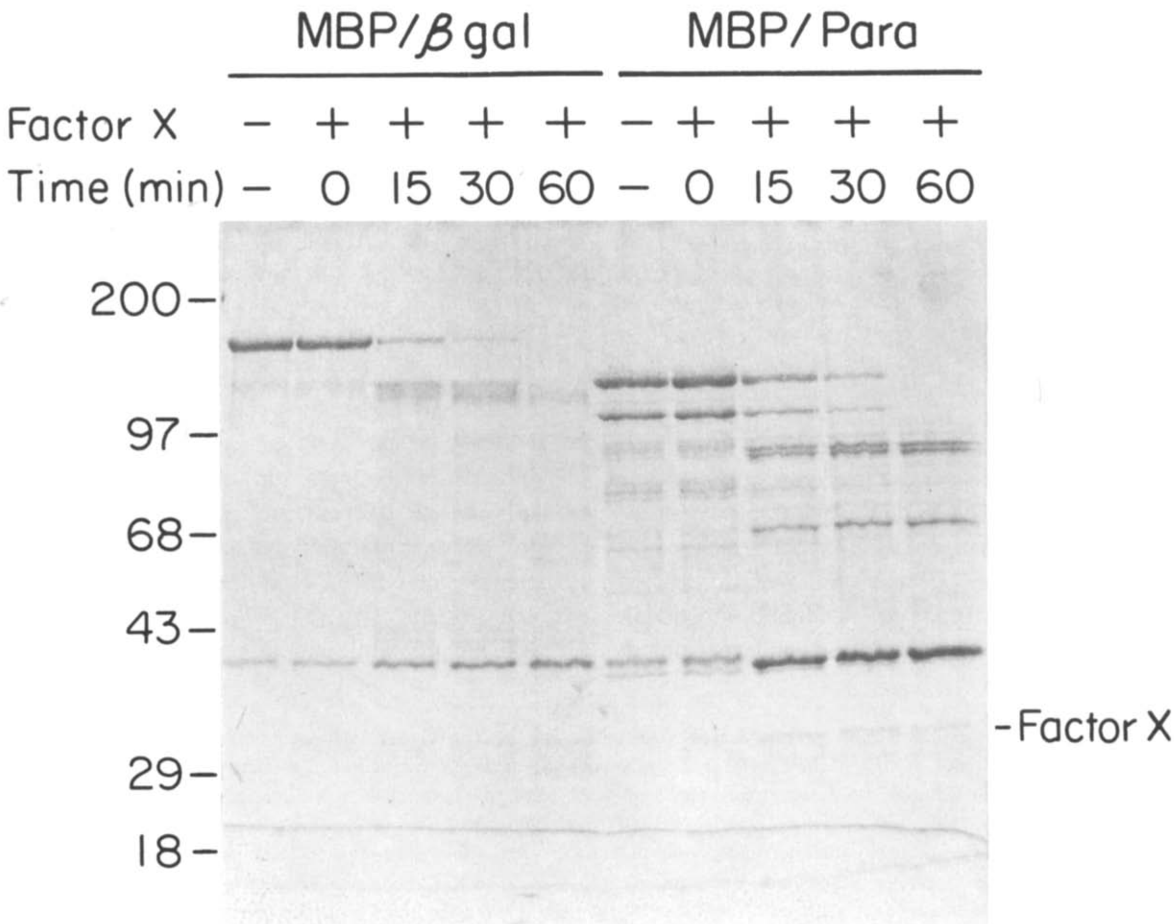

Fig. 4. Time course for digestion of fusion proteins with factor X_a. Protein samples were concentrated by ultrafiltration in Centricon microconcentrators (Amicon, Danvers, MA) to an approximate concentration of 0.5 mg/ml and then dialyzed extensively in 10 mM Tris · HCl pH 7.2. 200 ng of factor X_a were added to 20 μg each of denatured MBP-β-galactosidase and native MBP-paramyosin and digested at room temperature. 5 μg were removed at 0, 15, 30 and 60 min and reaction stopped by adding SDS-PAGE loading buffer and boiling for 2 min. The samples were then electrophoresed on a 0.1% SDS-10% polyacrylamide gel, and the gel stained after the run with Coomassie brilliant blue. The factor X_a protein band is shown. Proteins used as molecular weight markers (BRL 6001LA, bands not shown) were myosin heavy chain (200 kDa), phosphorylase B (97 kDa), bovine serum albumin (68 kDa), ovalbumin (43 kDa), carbonic anhydrase (29 kDa), and β-lactoglobulin (18 kDa). The degradation of the factor X_a-cleaved β-galactosidase is most likely due to a protease contaminating the MBP-β-galactosidase preparation since it is not seen in the digestion of MBP-paramyosin.

(d) Digestion of fusion proteins with factor X_a protease

The 4 amino acids (Ile-Glu-Gly-Arg) that constitute the recognition site for factor X_a protease lie between the two domains of the MBP-paramyosin and MBP-β-galactosidase fusion proteins. To separate the two domains we reacted each fusion protein with factor X_a, removed samples at various times and subjected them to SDS-PAGE (Fig. 4). Lanes 6 through 10 show the digestion of MBP-paramyosin with factor X_a. Over the period of the digestion the two upper bands of 126 and 111 kDa disappear, while two new lower bands of 95 and 71 kDa appear and the 40 kDa (MBP) band increases in intensity. In addition, the sum of the sizes of the cleaved bands equals the original, uncleaved band (within experimental error), demonstrating that the MBP-paramyosin fusion proteins are being cleaved into their two domains.

Similar results are seen for the MBP-β-galactosidase fusion protein in lanes 1 through 5. The 155-kDa protein disappears and a new protein of 125 kDa appears, while the 40-kDa band increases in intensity. In addition, the sum of the sizes of the cleaved bands equals the uncleaved band (again, within experimental error). The bottom band of 30 kDa seen in lanes 2 through 5 and 7 through 10 is due to the addition of factor X_a protease.

This interpretation of the cleavage products is confirmed by immunoblot analysis. Lanes 1 and 2 of Fig. 5 show intact and factor X_a-cleaved MBP-paramyosin respectively. The banding pattern (Fig. 5, left panel) is similar to that seen in Fig. 4. In both the intact and cleaved lanes, only the upper two bands are reactive with anti-*D. immitis* serum (Fig. 5, center panel) whereas only the lower band is reactive with anti-MBP serum (Fig. 5, right panel), demonstrating that cleavage with factor X_a separates the fusion protein into its two antigenic domains.

The MBP-paramyosin fusion is digested with factor X_a under native conditions, which suggests that the 4-aa recognition sequence for factor X_a is accessible to the protease in this large protein. The MBP-β-galactosidase fusion on the other hand, required denaturation with urea before factor X_a could cleave it. This suggests that the protease recognition site of the MBP-β-galactosidase fusion is inaccessible to the protease when the fusion is in its native state, but is made accessible by denaturation.

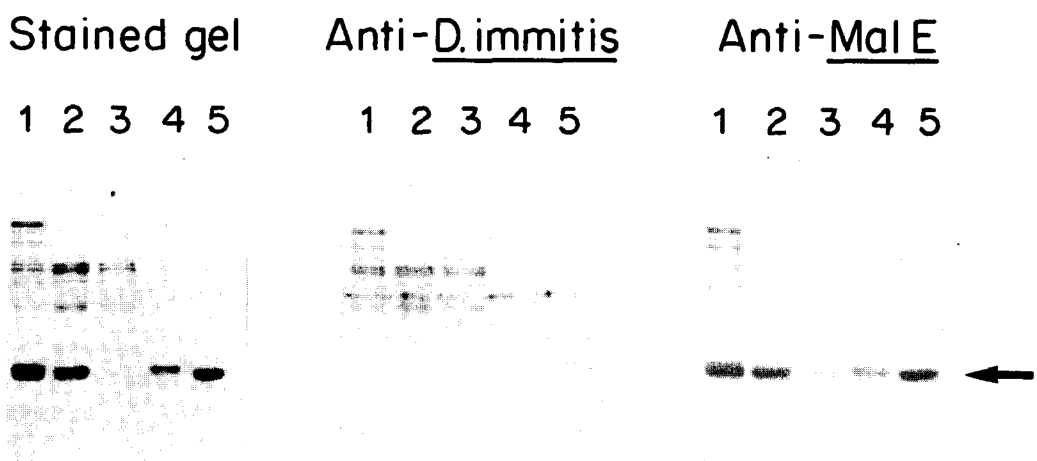

Fig. 5. Digestion of MBP-paramyosin with factor X_a and separation of protein domains. 1 mg of MBP-paramyosin was concentrated and dialyzed as described in Fig. 4 and then digested with 10 μg of factor X_a at room temperature for 16 h. The sample was checked for complete digestion by SDS-PAGE. The digested fusion protein was then adjusted to 1 M NaCl and passed over a 10 ml bed volume of crosslinked amylose resin. The column was washed with an equal volume of 10 mM Tris · HCl, pH 7.2, 1 M NaCl and these two fractions were pooled and concentrated as described above. The bound fraction was eluted with 10 mM maltose and concentrated, also as described (Guan et al., 1988). Lanes: 1, 5 μg of MBP-paramyosin; 2, 5 μg of MBP-paramyosin digested to completion with factor X_a; 3, 2 μg of flow-through of digested MBP-paramyosin from the crosslinked amylose column; 4, 2 μg of purified MBP domain eluted from the crosslinked amylose column; 5, 2 μg of MBP. Left panel, gel stained with Coomassie brilliant blue. Center panel, immuno blot using rabbit anti-*D. immitis* serum. Right panel, immuno blot using anti-MBP serum. Protein bands were visualized as in Fig. 3. Arrow points to MBP.

(e) Separation of fusion protein domains by affinity chromatography

The two, now cleaved, domains of the MBP-paramyosin fusion can be separated by affinity chromatography to the amylose resin. The MBP domain should still bind to the resin (after removal of maltose) while the paramyosin domain flows through. The cleaved MBP-paramyosin was passed once over the crosslinked amylose column. The flow-through fraction was collected and the MBP domain was then eluted as before with maltose. The results are shown in Fig. 5. Lanes 3 and 4 show the flow-through and eluted fractions of the factor X_a-cleaved MBP-paramyosin fusion. The upper two paramyosin bands are found in the flow-through fraction only. The MBP band is seen in the eluted fraction and only barely detectable levels are seen in the flow-through fraction. The center and left panels of Fig. 5 demonstrate again that the two upper bands, which now flow-through the amylose resin, are reactive with anti-*D. immitis* serum only, while the lower band, which does bind to the resin, is reactive with anti-MBP serum only. The yield of the paramyosin domain was approximately 20% of the fusion protein loaded onto the column (see section **f** below).

(f) Conclusions

The techniques described here are a simple and powerful approach to express and purify proteins in *E. coli* as fusions to MBP. Fusions to MBP are easy to construct and purification of the fusion protein gives high yield and purity in a single step carried out under non-denaturing conditions. In addition, the target protein domain can be separated from the MBP domain by cleavage with factor X_a protease whose 4 aa recognition sequence lies between the two domains. The target domain can then be separated from the MBP domain by affinity chromatography with amylose resin.

We have successfully demonstrated cleavage of the two domains with fusions to β-galactosidase from *E. coli* and paramyosin from *D. immitis*. The paramyosin domain of the MBP-paramyosin fusion was purified away from the MBP domain by re-passage over the crosslinked amylose column. The level of purity was such that barely detectable levels of the MBP domain were present in the purified paramyosin domain, as determined by immuno blot analysis. The yield of purified paramyosin domain, in this one experiment, was 20% of that loaded on the column. An alternative approach uses a Pharmacia Mono Q HR 5/5 or Q Sepharose HR 10/10 column in a Pharmacia FPLC System. The MBP domain elutes at 100 to 150 mM KCl and paramyosin domain elutes at 250 to 300 mM KCl in a Tris · HCl buffer of neutral pH. The separation is complete and recovery of the paramyosin domain is greater than 90%, as determined by SDS-PAGE. We are experimenting with other methods that will simplify the factor X_a cleavage, and increase the yields of the purified domains.

Plasmid pCG806fX has been modified to allow for α-complementation of insert DNA (pCG807fX). Several other vectors are being constructed that will allow DNA from cloning and expression vectors such as λgt11 to be easily fused to *malE*. All *malE* expression vectors, as well as information on their use, are available from New England Biolabs upon request.

We believe the ease of cloning and purification of fusion proteins that these *malE* fusion vectors allow will make them an important tool in protein research.

ACKNOWLEDGEMENTS

The encouragement and support of Donald Comb is greatly appreciated. The authors would also like to thank M. Philipp and T. Davis for anti-*D. immitis* serum; D. Landry for synthesis of oligodeoxynucleotides; and P. Arasu, R. Croft, R. Limberger, C. Poole, E. Rosenvold and I. Schildkraut for helpful discussions.

REFERENCES

Bedouelle, H. and Duplay, P.: Production in *Escherichia coli* and one-step purification of bifunctional hybrid proteins which bind maltose. Eur. J. Biochem. 171 (1988) 541–549.

Casadaban, M.J., Chou, J. and Cohen, S.N.: In vitro gene fusions that join an enzymatically active β-galactosidase segment to amino-terminal fragments of exogenous proteins: *Escherichia coli* plasmid vectors for the detection and cloning of translational initiation signals. J. Bacteriol. 143 (1980) 971–980.

Cohen, C., Szent-Györgyi, A.G. and Kendrick-Jones, J.: Paramyosin and the filaments of molluscan 'catch' muscles, I. Paramyosin: structure and assembly. J. Mol. Biol. 56 (1971) 223–237.

Guan, C.-d., Li, P., Riggs, P.D. and Inouye, H.: Vectors that facilitate the expression and purification of foreign peptides in *Escherichia coli* by fusion to maltose-binding protein. Gene 67 (1988) 21–30.

Germino, J. and Bastia, D.: Rapid purification of a cloned gene product by genetic fusion and site-specific proteolysis. Proc. Natl. Acad. Sci. USA 81 (1984) 4692–4696.

Haffey, M.L., Lehman, D. and Boger, J.: Site-specific cleavage of a fusion protein by renin. DNA 6 (1987) 565–571.

Jackson, C.M.: Factor X. In Spaet, T.H. (Ed.), Progress in Hemostasis and Thrombosis, Vol. 7. Grune and Stratton, New York, 1984, pp. 55–108.

Knight, D.H.: Heartworm heart disease. In Brandly, A., Cornelius, C.E. and Simpson, C.F. (Eds.), Advances in Veterinary Science and Comparative Medicine, vol. 21. Academic Press, New York, 1977, pp. 107–149.

Löwenadler, B., Nilsson, B., Abrahmsén, L., Moks, T., Ljundqvist, L., Holmgren, E., Paleus, S., Josephson, S., Philipson, L. and Uhlén, M.: Production of specific antibodies against protein A fusion proteins. EMBO J. 5 (1986) 2393–2398.

Löwenadler, B., Jansson, B., Paleus, S., Holmgren, E., Nilsson, B., Moks, T., Gunnar, P., Josephson, S., Philipson, L. and Uhlén, M.: A gene fusion system for generating antibodies against short peptides. Gene 58 (1987) 87–97.

Maniatis, T., Fritsch, E.F. and Sambrook, J.: Molecular Cloning. A Laboratory Manual. Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, 1982.

Nagai, K., Perutz, M.F. and Poyart, C.: Oxygen binding properties of human hemoglobins synthesized in *Escherichia coli*. Proc. Natl. Acad. Sci. USA 82 (1985) 7252–7255.

Nagai, K. and Thøgersen, H.C.: Generation of β-globin by sequence-specific proteolysis of a hybrid protein produced in *Escherichia coli*. Nature 309 (1984) 810–812.

Nilsson, B., Holmgren, E., Josephson, S., Gatenbeck, S., Philipson, L. and Uhlén, M.: Efficient secretion and purification of human insulin-like growth factor I with a gene fusion vector in Staphylococci. Nucleic Acids Res. 13 (1985) 1151–1162.

Philipp, M. and Davis, T.B.: Biochemical and immunologic characterization of a major surface antigen of *Dirofilaria immitis* infective larvae. J. Immunol. 136 (1986) 2621–2627.

Shuman, H.A., Silhavy, T.J. and Beckwith, J.R.: Labeling of proteins with β-galactosidase by gene fusion. J. Biol. Chem. 255 (1980) 168–174.

Silhavy, T.J., Berman, M.L. and Enquist, L.W.: Experiments with Gene Fusions. Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, 1984.

Slatko, B.E., Benner, J.S., Jager-Quinton, T., Moran, L.S., Simcox, T.G., Van Cott, E.M. and Wilson, G.G.: Cloning, sequencing and expression of the *TaqI* restriction-modification system. Nucleic Acids Res. 15 (1987) 9781–9796.

Valerie, K., Fronko, G., Long, W., Henderson, E.E., Nilsson, B., Uhlén, M. and de Riel, J.K.: Production and detection of coliphage T4 endonuclease V polyclonal and monoclonal antibodies using staphylococcal protein A hybrid proteins. Gene 58 (1987) 99–107.

Vermersch, P.S., Klass, M.R. and Bennett, G.N.: Use of DHFR-II fusion proteins to elicit specific antibodies. Gene 41 (1986) 289–297.

Winkelman, L.: Comparative studies of paramyosins. J. Biochem. Physiol. 55B (1976) 391–397.

Yanisch-Perron, C., Vieira, J. and Messing, J.: Improved M13 phage cloning vectors and host strains: nucleotide sequences of the M13mp18 and pUC19 vectors. Gene 33 (1985) 103–119.

Young, J., Hockmeyer, W., Gross, M., Ballou, R., Wirtz, R., Trosper, J., Beaudoin, R., Hollingdale, M., Miller, L., Diggs, C. and Rosenberg, M.: Expression of *Plasmodium falciparum* circumsporozoite proteins in *Escherichia coli* for potential use in human malaria vaccine. Science 228 (1985) 958–962.

Young, R.A. and Davis, R.W.: Efficient isolation of genes by using antibody probes. Proc. Natl. Acad. Sci. USA 80 (1983) 1194–1198.

Communicated by F. Barany.